

# Week 2: Probability Fundamentals

STA237: Probability, Statistics, and Data Analysis I

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Week of Sept. 8, 2025

# Learning Outcomes

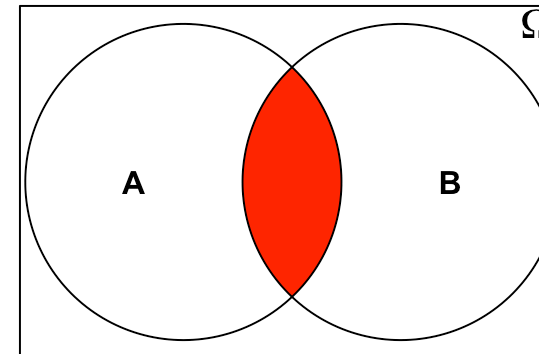
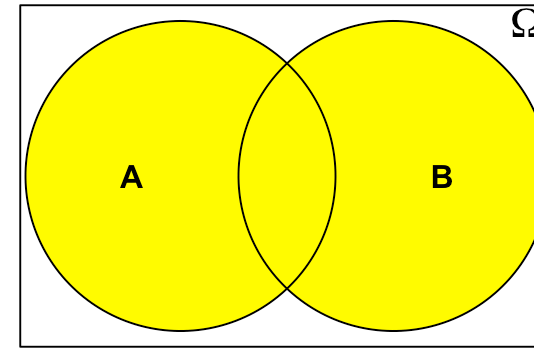
By the end of this week, you will have learned to

- ☐ Use set notation and Venn diagrams to represent events in a given sample space
- ☐ Distinguish between mutually exclusive (i.e., disjoint) and independent events
- ☐ Explain what is meant and described by probability models
- ☐ State the three probability axioms
- ☐ Use axioms to relate probabilities of sets of events (e.g.,  $P(A \cup B)$ ,  $P(A)$ ,  $P(B)$ ,  $P(A^c)$ )
- ☐ Apply inclusion-exclusion principle ('Addition Rule') to find probabilities involving unions or intersections of events
- ☐ State and explain the result of Weak Law of Large Numbers
- ☐ State and use the fundamental principle of counting

# Last week

We introduced the following terminology:

- ✓ Random experiment
- ✓ Sample space ( $\Omega, S$ )
- ✓ Event - simple and compound (e.g.,  $A$ )
- ✓ Complementary event (e.g.,  $A^c, \overline{A}, A'$ )
- ✓ Union of events (e.g.,  $A \cup B$ )
- ✓ Intersection of events (e.g.,  $A \cap B$ )



# Example 1 - Event Operator

Let's keep things simple. Suppose we roll two dice and record the pairs of outcomes. List the following events:

- a. The sample space
- b. The event that doubles were rolled.
- c. The event that the outcome sums to 8.
- d. The event where one die has twice the face value of the other.

# Event Relations

Consider two events  $A$  and  $B$  in the sample space  $\Omega$ :

- Two events are **independent** if the occurrence of one event does not alter the chance that the other event occurs.
- Two events are **dependent** if the occurrence of one event alters the chance that the other event occurs.
- Two events are **mutually exclusive** or **disjoint** if events  $A$  and  $B$  cannot occur together or simultaneously as an outcome of a random experiment

# Independence vs Mutually Exclusive

It's common to mix these two terms up because they *seem* similar!

## Independence

Whether event  $A$  occurs or not doesn't impact the chances of  $B$  occurring.

- If  $A$  occurs, then the chance of  $B$  is \_\_\_\_\_
- If  $A$  does not occur, then the chance of  $B$  is \_\_\_\_\_

E.g., Let  $\Omega$  be the sample space for the result of two consecutive coin tosses. The sample space would be  $\Omega = \{\text{_____}\}$ . Let:

- $A$  be the event that the coin landed heads on the **first** toss
- $B$  be the event that the coin landed tails on the **second** toss

## Mutually Exclusive

Whether event  $A$  occurs or not directly impacts the chances of  $B$  occurring

- If  $A$  occurs, then the chance of  $B$  is \_\_\_\_\_
- If  $A$  does not occur, then the chance of  $B$  is \_\_\_\_\_

E.g., Let  $\Omega$  be the sample space for the result of two consecutive coin tosses. The sample space would be  $\Omega = \{\text{_____}\}$ . Let:

- $A$  be the event that the coin landed heads on the **first** toss
- $B$  be the event that the coin landed tails on the **first** toss

Mutual exclusivity is a special case of

\_\_\_\_\_.

## Example 2 - Event Relations

Determine whether the following pairs of events or outcomes are mutually exclusive, independent, or dependent. Opt for the most precise label where two answers are possible.

1. A student at U of T is randomly selected and surveyed about their course work. Let  $A = \{\text{Enrolled in 100 level course}\}$  and  $B = \{\text{Enrolled in 300 level course}\}$
2. A card is drawn at random from a deck of 52 playing cards. Let  $A = \{\text{Black card}\}$  and  $B = \{\text{Ace of Spades}\}$ .
3. You and your friend flip a coin three times. Let  $A = \{\text{You land HHH}\}$  and  $B = \{\text{Friend lands TTT}\}$

# Commutative, Associative, Distributive Laws

Unions and intersections of events have some useful properties. Consider a sample space  $\Omega$  with three events  $A, B, C$ :

- **Commutative Law:**  $A \cup B = B \cup A$
- **Associative Law:**  $(A \cup B) \cup C = A \cup (B \cup C)$  and  $(A \cap B) \cap C = A \cap (B \cap C)$
- **Distributive Law:**  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Verify these properties using a Venn diagram!



# DeMorgan's Laws

For two events  $A$  and  $B$ :

- $(A \cup B)^c = A^c \cap B^c$
- $(A \cap B)^c = A^c \cup B^c$

This can be extended more generally to the set of events  $\{A_1, A_2, \dots, A_n\}$ :

- $(\bigcup_{i=1}^n A_i)^c = \bigcap_{i=1}^n A_i^c$
- $(\bigcap_{i=1}^n A_i)^c = \bigcup_{i=1}^n A_i^c$

DeMorgan's Laws can be very helpful in re-expressing event operators into more familiar forms while problem solve or in proofs of event relations.

# Probability Axioms

## Probability Function

In a random experiment with sample space  $\Omega$ , a **probability function**  $P$  is a function on  $\Omega$  that assigns to each element  $\omega$  in the sample space a numerical value that measures the *chance* that event  $\omega$  will occur. This is denoted as  $P(\omega)$ . A probability function has the following properties:

1.  $P(\omega) \geq 0$

for all  $\omega \in \Omega$

2. 
$$\sum_{\omega \in \Omega} P(\omega) = 1$$

3. For all events  $A \subseteq \Omega$ ,

$$P(A) = \sum_{\omega \in A} P(\omega)$$

# Probability and Event Relations

All probability models have the three properties, or “probability axioms” listed on the previous slide. Using these axioms, we can show how probabilities of events are related!

## Probability of Complementary Event

Intuitively, for a given event  $A$  with probability  $P(A)$ , we would expect that the probability of its complement  $A^c$  will be  $P(A^c) = \underline{\hspace{2cm}}$ . Let's show this:

## Example 3 - Probability and Event Relations

In a group of 15 friends, 10 had watched KPop Demon Hunters, 6 listen to K-pop regularly, and 3 have neither watched Kpop Demon Hunters nor regularly listen to K-pop. Try to represent this information in a Venn diagram.

Can you determine how many friends have watched KPop Demon Hunters AND also regularly listen to K-pop?

# Probability and Event Relations

Example 3 is a classic example of the **Inclusion-Exclusion** principle, sometimes called the Addition Rule, which tells us for two events  $A$  and  $B$  in a sample space:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Let's show this using probability axioms:

## Example 4 - Inclusion-Exclusion Principle

Among a group of 60 students, 28 could roll their tongue and 15 have attached earlobes. 8 students could both roll their tongue and have attached earlobes. We conduct a random experiment where a student is sampled at random and we note whether they could roll their tongue or if they have attached earlobes. Let

- $T$  be the event that the student can roll their tongue.
- $E$  be the event that the student has attached earlobes.

Represent the following events in notation, and determine the number of elements in each event:

a. The student cannot roll their tongue.

b. The student can neither roll their tongue nor has attached earlobes.

# Progress Check

By the end of this week, you will have learned to

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- ✓ State the three probability axioms
- ✓ Use axioms to relate probabilities of sets of events (e.g.,  $P(A \cup B)$ ,  $P(A)$ ,  $P(B)$ ,  $P(A^c)$ )
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- ☐ State and explain the result of Weak Law of Large Numbers
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# Inclusion-Exclusion Principle: General Case

We can extend the inclusion-exclusion principle beyond two events to some set of  $n$  events:  $\{A_1, A_2, \dots, A_n\}$ . It becomes more complex because we need to account for the intersection of  $r$  subsets of events:

## General Addition Rule

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_{i=1}^n P(A_i) - \sum_{i_1 < i_2} P(A_{i_1} \cap A_{i_2}) + \dots \\ &\quad + (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_r}) + \dots \\ &\quad + (-1)^{n+1} P(A_1 \cap A_2 \cap \dots \cap A_n) \end{aligned}$$

*Note:* If the set of events  $\{A_1, A_2, \dots, A_n\}$  are **disjoint**, then  $P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$

Let's consider for three events  $\{A_1, A_2, A_3\}$ :



## Example 5 - Addition Rule

**MMSA Ex. 15** A consulting firm presently has bids out on three projects. Let  $A_i = \{Awardeprojecti\}$  for  $i = 1, 2, 3$  and suppose that:

- $P(A_1) = 0.22$
- $P(A_2) = 0.25$
- $P(A_3) = 0.28$
- $P(A_1 \cap A_2) = 0.11$
- $P(A_1 \cap A_3) = 0.05$
- $P(A_2 \cap A_3) = 0.07$
- $P(A_1 \cap A_2 \cap A_3) = 0.01$

Describe what each event represents and find their probabiity:

a.  $A_1^c \cap A_2^c$

## Example 5 - Addition Rule

**MMSA Ex. 15** A consulting firm presently has bids out on three projects. Let  $A_i = \{\text{Awarded project } i\}$  for  $i = 1, 2, 3$  and suppose that:

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- $P(A_2 \cap A_3) = 0.07$
- $P(A_1 \cap A_2 \cap A_3) = 0.01$

Describe what each event represents and find their probability:

b. How would you represent the event where they are awarded at least two projects? Find this probability.

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- ✓ State and explain the result of Weak Law of Large Numbers
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# Counting and Probability

How do we find the probability of an event occurring?

A common interpretation of probability is one that defines it as the event's long-run relative frequency of occurrence. This means the proportion of times an event occurs in  $n$  repeated trials of the random experiment, and observing the limiting behaviour of this proportion as  $n \rightarrow \infty$ !

## Probability and Relative Frequency

Consider the sample space of a random experiment that consists of **equally likely** elements. The probability of some event  $A \subseteq \Omega$  is the relative frequency of event  $A$  in  $\Omega$ :

$$P(A) = \frac{n(A)}{n(\Omega)}$$

Where  $n(\cdot)$  denotes the number of elements in  $(\cdot)$  This probability definition holds true **only if** each element in  $\Omega$  is **equally likely**!

# Weak Law of Large Numbers

In probability theory, the **Weak Law of Large Numbers** states that as the number of independent trials increases, the observed average result will converge to the theoretical average.

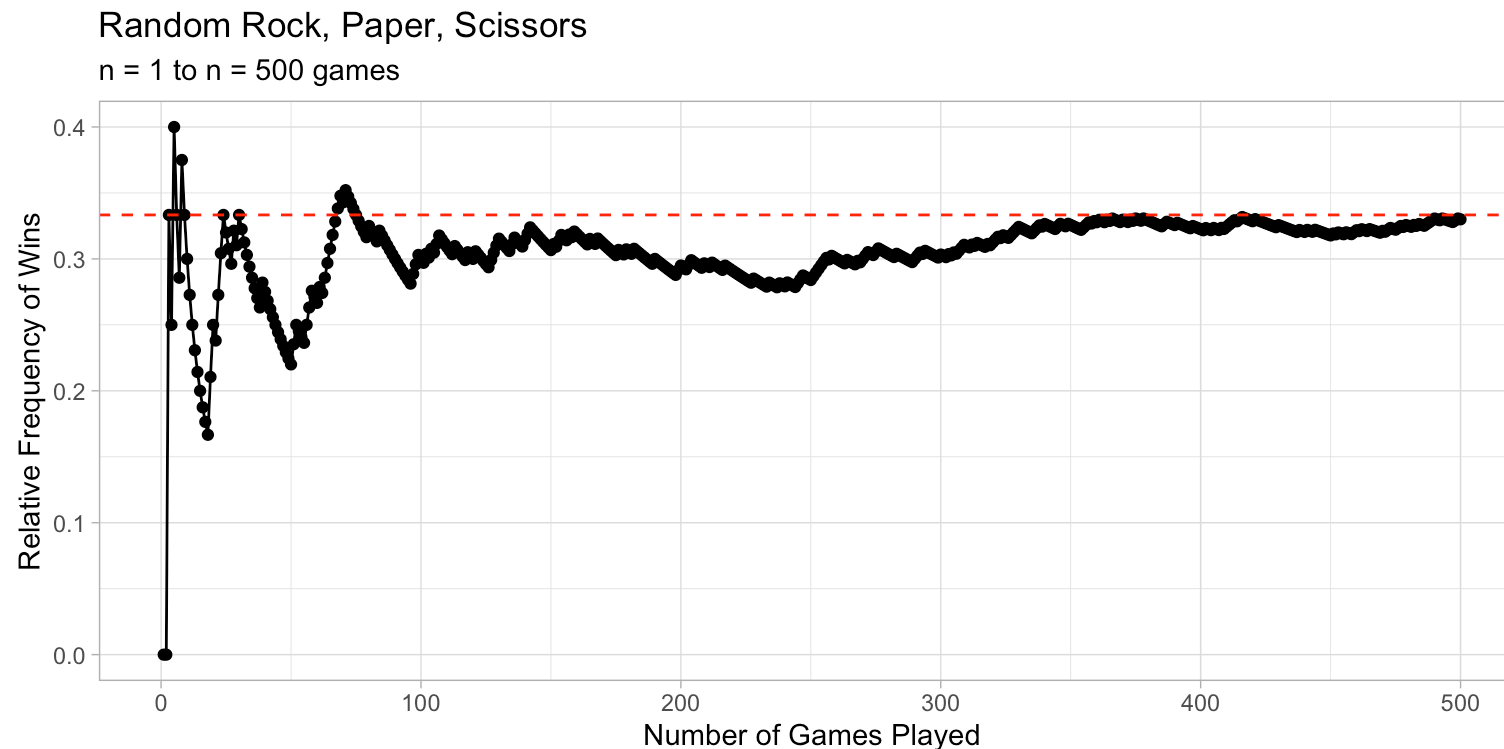
When we repeat a random experiment, the average times we observe some event  $A$  is the relative frequency of  $A$ . As we increase the number of times we repeat the same random experiment, the relative frequency of  $A$  will converge to its theoretical relative frequency, which is its probability.

# Long-Run Relative Frequency

Consider the random experiment where you play rock-paper-scissors against a machine that randomly plays one of rock, paper, or scissors with equal probability. Intuitively, we would think that we have a 1 in 3 chance of winning.

This doesn't mean we are guaranteed to win once every three games, but in the long-run, we would expect to win, on average, one out of every 3 games.

Let's simulate the results of 500 games of rock, paper, scissors:



# Counting and Probability

Having counting strategies can help us find probabilities of events when working with sample spaces that consist of equally-likely outcomes.

- We can apply inclusion-exclusion principle to counts to count the number of elements in unions of events:  
$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$
- To count the number of elements in a random experiment that involve multiple **ordered** stages, we can use the **Fundamental Principle of Counting**

## Fundamental Principle of Counting

Consider an  $m$  element sequence  $(a_1, a_2, \dots, a_m)$ . If there are  $n_1$  unique possible values for the first element,  $n_2$  unique possible outcomes for the second element, ...,  $n_m$  unique possible outcomes for the  $m^{th}$  element, then the total number of unique sequences is:

$$\prod_{i=1}^m n_i$$

# Toy Example

A sandwich stand allows you to customize your sandwich with ingredients of your choice. Their default sandwich build is to layer first greens, followed by deli meat, and finished with your cheese of choice. There are **three greens** options, **five deli meat** options, and **four cheese** options. How many unique sandwiches can be created with these options if a customer can only select one ingredient for each layer?

How does this number change if the ingredients can be layered in any order?



## Example 6

When opening a credit or debit account, you are required to set a 4 to 6 digit pin in order to access the account in the future.

- a. How many unique PIN exists if all digits must be numeric?
  
  
  
  
  
  
  
  
  
  
- b. How many unique PIN exists if all digits are numeric and can only be used ONCE?

# Try!

How many different divisors are there for the number 67, 500?

*Hint: Start by expressing as a product of primes*

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